CLASS 10 — MATHS · FORMULA SHEET

Chapter 5: **Arithmetic Progressions** | Quick revision, one page

Arithmetic progression

- A list $a,\ a+d,\ a+2d,\ldots$ with first term a and common difference $d = a_{k+1} - a_k$ (constant).
- d can be positive, negative or zero.

nth term

- $a_n = a + (n-1)d$.
- nth term from the end of a finite AP with last term l: $l - (n-1)d$.
- Check for AP: $a_{k+1} - a_k$ is the same for all k.

Sum of first n terms

- $S_n = \frac{n}{2}\,[2a + (n-1)d]$.
- $S_n = \frac{n}{2}\,(a + l)$ when the last term l is known.
- $a_n = S_n - S_{n-1}$.
- Sum of first n natural numbers $= \frac{n(n+1)}{2}$; first n odd numbers $= n^2$.

Useful results

- Three consecutive terms: $a-d,\ a,\ a+d$. Four terms: $a-3d,\ a-d,\ a+d,\ a+3d$.
- If a, b, c are in AP then $2b = a + c$.
- Number of terms of $a,\ldots,l$: $n = \frac{l-a}{d} + 1$.

Word problems

- Decide the first term, d and what is asked (a term or a sum); reject non-integer or negative n.

MASTERMIND